KAIZEN MATHS

Lesson starter

Column Vectors and Vector Geometry

Scalar Multiples

Name _____

S1

$\mathrm{a} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$

Find 4a.

S2

$\mathrm{a} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

Find 2a - 1b.

S3

$\mathrm{a} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$

Find -3a.

S4

$\mathrm{a} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$

Find -2a + 3b.

Show your working

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Show your working

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Exit ticket

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EA1

$$\mathrm{a} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

Find -2a.

EA2

$$\mathrm{a} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\mathrm{b} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

Find -2a - 1b.

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Column Vectors and Vector Geometry

Scalar Multiples

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$$\mathrm{a} = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

Find -2a.

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$$\mathrm{a} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

Find -2a.

EA2

$$\mathrm{a} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\mathrm{b} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

Find -2a - 1b.

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Show your working

KAIZEN MATHS

Exit ticket

Column Vectors and Vector Geometry

Scalar Multiples

Name _____

EB1

$$\mathrm{a} = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

Find -2a.

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Show your working

Teacher answers & working

Column Vectors and Vector Geometry · Scalar Multiples

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S1

$\mathrm{a} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$

Find 4a.

Answer:
$\begin{pmatrix} -16 \\ -4 \end{pmatrix}$

Working:

Multiply every component by 4.

$4\mathrm{a} = \begin{pmatrix} 4 \times -4 \\ 4 \times -1 \end{pmatrix}$

$4\mathrm{a} = \begin{pmatrix} -16 \\ -4 \end{pmatrix}$

S2

$\mathrm{a} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

Find 2a - 1b.

Answer:
$\begin{pmatrix} -6 \\ 9 \end{pmatrix}$

Working:

Work component by component.

$2\mathrm{a} - \mathrm{b} = \begin{pmatrix} 2(-1) - 1(4) \\ 2(3) - 1(-3) \end{pmatrix}$

$= \begin{pmatrix} -6 \\ 9 \end{pmatrix}$

Teacher answers & working

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S3

$\mathrm{a} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$

Find -3a.

Answer:

$\begin{pmatrix} 9 \\ -6 \end{pmatrix}$

Working:

Multiply every component by -3.

$-3\mathrm{a} = \begin{pmatrix} -3 \times -3 \\ -3 \times 2 \end{pmatrix}$

$-3\mathrm{a} = \begin{pmatrix} 9 \\ -6 \end{pmatrix}$

S4

$\mathrm{a} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$

Find -2a + 3b.

Answer:

$\begin{pmatrix} 20 \\ -5 \end{pmatrix}$

Working:

Work component by component.

$-2\mathrm{a} + 3\mathrm{b} = \begin{pmatrix} -2(-4) + 3(4) \\ -2(1) + 3(-1) \end{pmatrix}$

$= \begin{pmatrix} 20 \\ -5 \end{pmatrix}$

Teacher answers & working

Column Vectors and Vector Geometry · Scalar Multiples

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EA1

$\mathrm{a} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$

Find -2a.

Answer:

$\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

Working:

Multiply every component by -2.

$-2\mathrm{a} = \begin{pmatrix} -2 \times -2 \\ -2 \times -3 \end{pmatrix}$

$-2\mathrm{a} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$

EA2

$\mathrm{a} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$

Find -2a - 1b.

Answer:

$\begin{pmatrix} -5 \\ -5 \end{pmatrix}$

Working:

Work component by component.

$-2\mathrm{a} - \mathrm{b} = \begin{pmatrix} -2(4) - 1(-3) \\ -2(1) - 1(3) \end{pmatrix}$

$= \begin{pmatrix} -5 \\ -5 \end{pmatrix}$

Teacher answers & working

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EB1

$a = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$

Find -2a.

Answer:

$\begin{pmatrix} 4 \\ -4 \end{pmatrix}$

Working:

Multiply every component by -2.

$-2a = \begin{pmatrix} -2 \times -2 \\ -2 \times 2 \end{pmatrix}$

$-2a = \begin{pmatrix} 4 \\ -4 \end{pmatrix}$